

| W-2. Drinking Water Spring Improvement                                                                               |                                                                                                 |                                                                                               |                                                                                    |                                                                                     |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  <p>Water Resources Development</p> |  <p>Medium</p> |  <p>High</p> |  |  |
| Measure Type                                                                                                         | Cost                                                                                            | Preferences by Community                                                                      | Hazards                                                                            | Implementation Time of Measures                                                     |

| Descriptions                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |           |      |            |      |            |      |           |
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| Measures                       | <p>A fundamental human right, access to safe drinking water is also vital for public health, especially in excluded and rural areas. Drinking water springs are essential sources of drinkable water for residents in many places, including Bangladesh's hilly regions. However, pollution, climate change, and environmental degradation are posing a growing threat to these springs. Ensuring sustainable access to clean water requires putting into practice efficient conservation techniques for improving drinking water springs.</p>                                                          |      |           |      |            |      |            |      |           |
|                                | <p>Enhancing the CHT' drinking water springs is an essential conservation strategy that can improve local communities' livelihoods and public health. We can provide long-term solutions to the area's water problems by prioritizing comprehensive site assessment, implementing protective measures in place, improving water quality, and encouraging community involvement. In the long run, these initiatives will support a healthier and more sustainable future for the CHT by ensuring access to safe drinking water and strengthening people' overall resilience to environmental change.</p> |      |           |      |            |      |            |      |           |
| Location                       | R-114                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R-99 | R-65      | K-98 | K-76       | K-71 | B-173      | B-77 | B-76      |
|                                | ■                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ■    | ■         | ■    | ■          | ■    | ■          | ■    | ■         |
| Applicable Sites               | <ul style="list-style-type: none"><li>• Natural Springs in Upland Areas</li><li>• Forested Catchment Zones</li><li>• Valley Springs &amp; Low-Lying Recharge Areas</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                            |      |           |      |            |      |            |      |           |
| Slope                          | Less than 3%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      | 3% to 15% |      | 15% to 30% |      | 30% to 60% |      | Above 60% |
|                                | √                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      | √         |      |            |      |            |      |           |
| Design options and performance | <ul style="list-style-type: none"><li>• Establish buffer areas around springs to prevent contamination.</li><li>• Use sand filters, bio-fences, and sedimentation tanks for water purification.</li><li>• Suitable for transporting water from upland sources to communities with minimal infrastructure.</li><li>• Construction of small reservoirs, tanks, or recharge ponds for year-round availability.</li></ul>                                                                                                                                                                                   |      |           |      |            |      |            |      |           |
| Feasibility criteria           | <p><b>Technical Feasibility</b></p> <ul style="list-style-type: none"><li>• Ensures slope stability, proper filtration, and sustainable water extraction.</li><li>• Requires monitoring systems for water quality and flow rate.</li></ul> <p><b>Environmental Feasibility</b></p> <ul style="list-style-type: none"><li>• Prevents sedimentation and erosion around the spring.</li></ul>                                                                                                                                                                                                              |      |           |      |            |      |            |      |           |

- Natural Springs in Upland Areas
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- Environmental Feasibility**
- Prevents sedimentation and erosion around the spring.

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|                                  | <ul style="list-style-type: none"> <li>• Maintains natural hydrological balance.</li> <li>• Encourages efficient recharge and controlled extraction.</li> </ul> <p><b>Economic Feasibility</b></p> <ul style="list-style-type: none"> <li>• Uses locally available materials for construction.</li> <li>• Improves community health and water security.</li> <li>• Promotes community-led management to minimize operational expenses.</li> </ul> <p><b>Social Feasibility</b></p> <ul style="list-style-type: none"> <li>• Engages local people in protection, maintenance, and management.</li> <li>• Ensures reliable access to safe drinking water, reducing health risks.</li> <li>• Aligns with government policies and development programs.</li> </ul> |            |               |             |
| <b>Operation and maintenance</b> | <ul style="list-style-type: none"> <li>• <b>Routine Maintenance</b> –Vegetation control, and erosion repair.</li> <li>• <b>Seasonal Adjustments</b> – Rainwater harvesting in the monsoon, flow regulation in the dry season.</li> <li>• <b>Community Involvement</b> – Local user groups, support from NGOs and government agencies.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
| <b>Cost</b>                      | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
| <b>Benefits</b>                  | Benefits are as follows-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Canopy Cover Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
|                                  | Carbon Sequestration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Crop Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                                  | Crop Failure Risk Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Ensuring Food Security                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                                  | Fruit Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | Forestry Goods Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Livestock Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Fisheries Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Alternative Livelihood Generation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | Increase of Income                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Soil Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Improved Soil Health                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Soil Fertility Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Sustainable Water Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                                  | Water Quality Improvement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | Water Purification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | River Bank Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Surface Runoff Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                                  | Groundwater Recharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Pest and Disease Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Reduction of Pesticides and Fertilizer Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Ecosystem and Biodiversity Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |

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|                                  | Better Communications                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Better Education                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | Better WASH facilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
| <b>Implementation Challenges</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Land unavailable/high land area required                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Supply chain disconnection                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | Land right related problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | High human investment capital need                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | Insufficient infrastructures                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Less demand of agroforestry product                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Environmentally unsustainable                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Socially unacceptable                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Less beneficial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
| <b>Environmental performance</b> | <ul style="list-style-type: none"> <li>• Soil Erosion Control – Reduces sediment accumulation in water sources.</li> <li>• Water Conservation – Prevents excessive runoff, promoting groundwater recharge.</li> <li>• Ecosystem Protection – Maintains natural vegetation and biodiversity.</li> <li>• Climate Resilience – Enhances adaptability to droughts and rainfall variability.</li> </ul>                                                                                              |            |               |             |
| <b>Remarks</b>                   | <p>Success in water resource management depends on active community participation and well-structured training programs that empower local stakeholders. Sustainable water governance requires coordination with government policies, NGOs, and research institutions to develop comprehensive and long-lasting solutions. Additionally, drinking water conservation should be integrated into a broader watershed management plan to support landscape restoration and ecological balance.</p> |            |               |             |